

WMDR2-core

Package in package 'Model'

jkl created on 20

WMDR2-core diagram

Class diagram in package 'WMDR2-core'

WMDR2core

Notes

1. temporalExtent shall be realized as time:interval
2. temporal<Name> elements other than temporalExtent shall be realized as a dictionary of times, one for the values of <Name>, e.g., temporalTerritory = {
"interval": [2000-01-01, 2024-07-13],
"territory": [<ISO1>, "<ISO2>"]
}.
3. Following the OGC Features model, geometry = geolocation = coordinates + elevation

v0.1.4 Introduce instrument class.

v0.1.5 Expand instrument class.

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Model Report

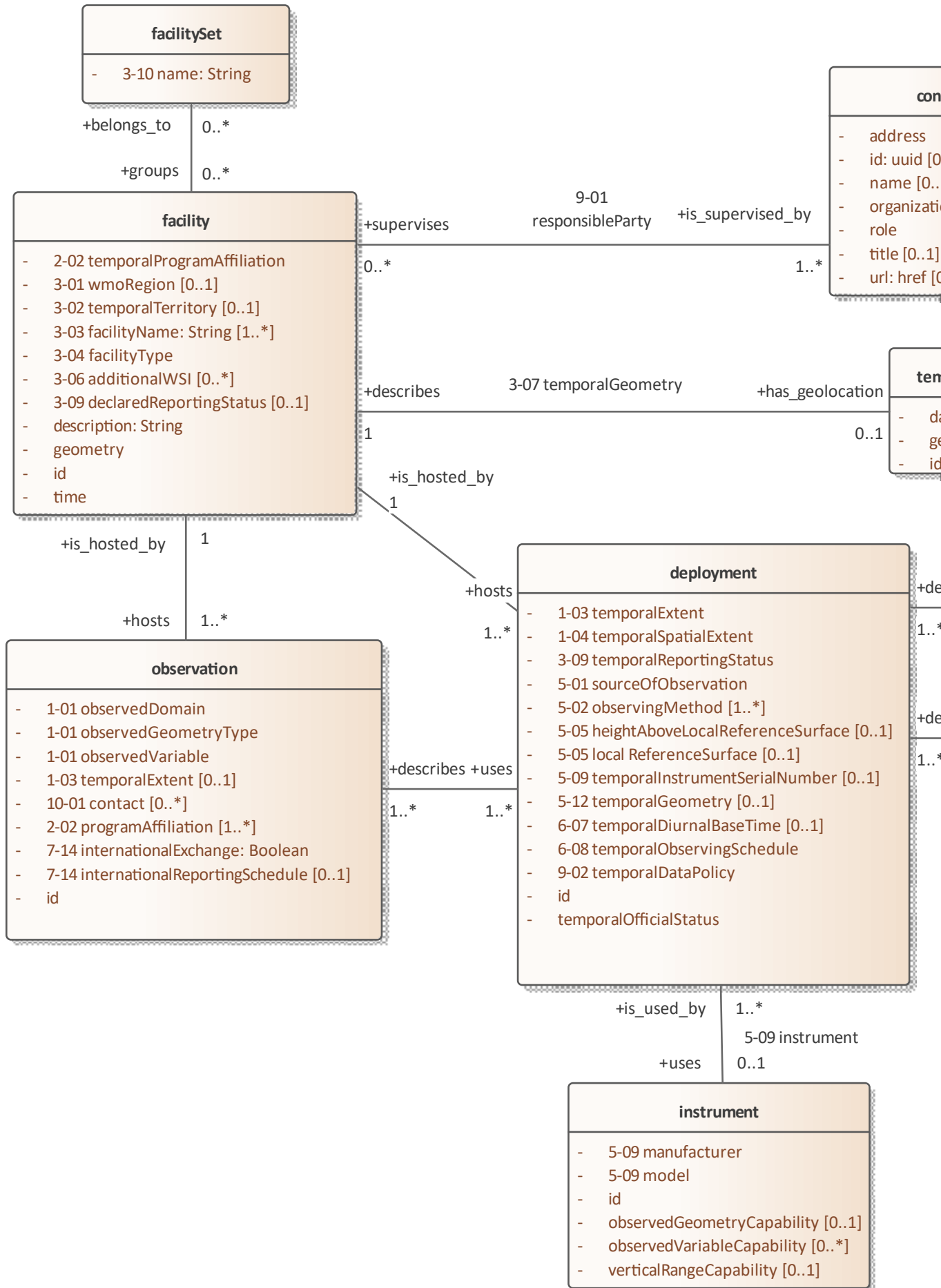


Figure 1: WMDR2-core

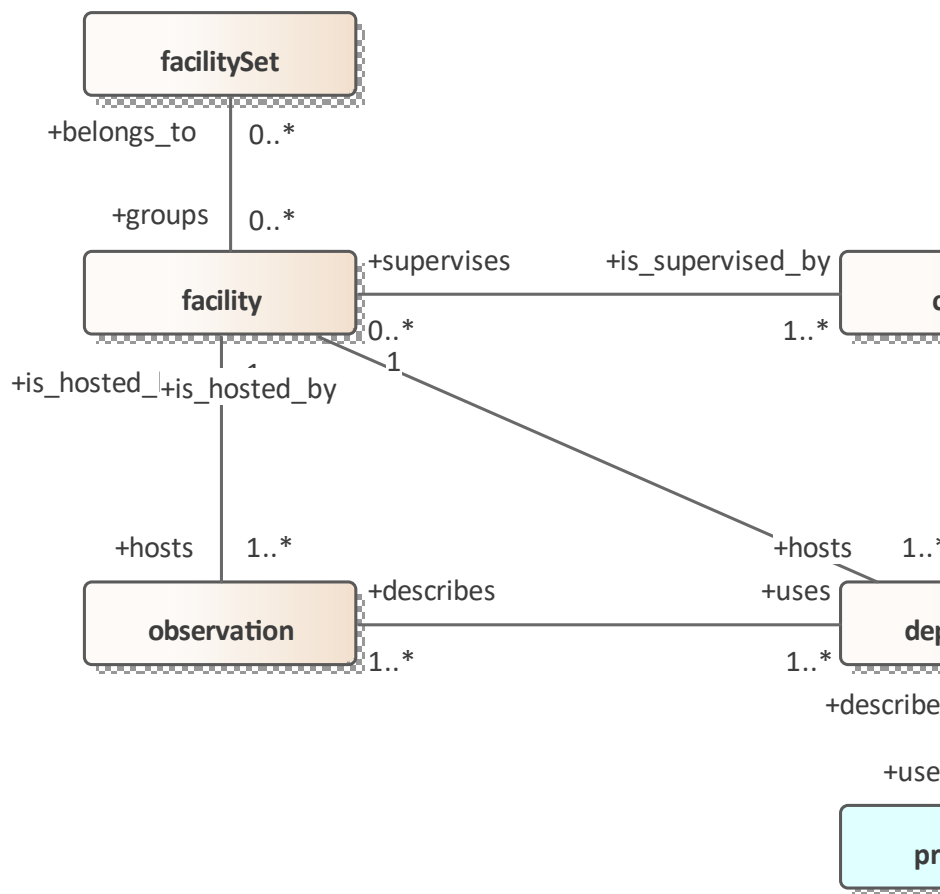


Figure 2: WMDR2-core-logical

contact

Class in package 'WMDR2-core'

impkement according to VCard 4.0 specification [0].

[0] <https://datatracker.ietf.org/doc/html/rfc6350#page-23>

ATTRIBUTES	
	address : Private

ATTRIBUTES **id : uuid Private**

Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

[Is static]

 **name : Private**

Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

Either a full name, or a sequence of name parts. Examples:

Dr. Jörg Klausen (this corresponds to FN in vcard)

Klausen;Jörg;Dr.; (this corresponds to N in vcard)

[Is static]

 **organization : Private**

Example:

ORG:Federal Office of Meteorology and Climatology MeteoSwiss

[Is static]

 **role : Private**

[Is static]

 **title : Private**

Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

[Is static]

 **url : href Private**

Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

[Is static]

ASSOCIATIONS Association (direction: Unspecified) 9-01 responsibleParty

specific instrument is rotated in and out, for example as part of operational maintenance

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ATTRIBUTES

 1-03 temporalExtent : Private
Alias: records:interval

[Is static

 1-04 temporalSpatialExtent : Private
Alias: records:geometry

Typical spatial geo-referenced volume covered by the observations.

EXAMPLES:

- (a) Air temperature at a surface observing site: Sydney Airport, New South Wales, Australia alt. 6.0 m above mean sea level (amsl);
- (b) The projected area or volume of the cone around a particular weather radar with a maximum reflectivity) and 150 km (Doppler); to be expressed as a geometric shape;
- (c) Three-dimensional grid of radar pixels;
- (d) For infrared and visible imagery by meteorological satellite (sun-synchronous): global;
- (e) For nadir sounding: atmospheric column above ocean;
- (f) The Japan Meteorological Agency Advanced Himawari Imager has a 10 min cycle full resolution) located above 140.68°E geostationary orbit at an altitude of 35,786 km;
- (g) River discharge by gauge: size and geometric shape of a river catchment.

Q: what is the spatial extent for a)?

[Is static

 3-09 temporalReportingStatus : Private

[Is static

 5-01 sourceOfObservation : Private

[Is static

ATTRIBUTES

Linked to heightAboveReferenceSurface

[Is static

 5-09 temporalInstrumentSerialNumber : Private
Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

Implemented as

```
"temporalInstrumentSerialNumber": {  
  "serialNumber": [...],  
  "datetimes": [...],  
}
```

[Is static

 5-12 temporalGeometry : Private
Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

- Geospatial location of instrument/sensor

[Is static

 6-07 temporalDiurnalBaseTime : Private
Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

indicates the end (hour) of a 24-hour period. Mandatory for 24-hour periodic observation precipitation

[Is static

 6-08 temporalObservingSchedule : Private

complex type: when are observations made? Should be expressed as a temporal coverage from 06-18 UTC"

[Is static

 9-02 temporalDataPolicy : Private

remove from WMDR2 and leave for WCMP2?

ASSOCIATIONS



Association (direction: Unspecified)

Source: Public describes (Class) deployment

Cardinality: [1..*]

Target: Public

Cardinality

ObservingProc



Association (direction: Unspecified) 5-09 instrument

Source: Public is_used_by (Class) deployment

Cardinality: [1..*]

Target: Public

Cardinality



Association (direction: Unspecified) 7-03 temporalReportingSchedule

Source: Public describes (Class) deployment

Cardinality: [1..*]

Target: Public

temporalRepo

Cardinality



Association (direction: Unspecified)

Source: Public is_hosted_by (Class) facility

Cardinality: [1]

Target: Public

Cardinality



Association (direction: Unspecified)

Source: Public describes (Class) observation

Cardinality: [1..*]

Target: Public

Cardinality

facility

Class in package 'WMDR2-core'

ATTRIBUTES	
	[Is static]
 3-03 facilityName : String Private Multiplicity: ([1..*], Allow duplicates: 0, Is ordered: False)	[Is static]
 3-04 facilityType : Private	[Is static]
 3-06 additionalWSI : Private Alias: externalIds Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) An array of well-formed WSIs	[Is static]
 3-09 declaredReportingStatus : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) Condition: must be provided if no deployment is specified; otherwise: Optional, inferred temporalReportingStatus + temporalProgramAffiliation. Exception: "closed" over-rules deployment, in which case all deployments should be terminated	[Is static]
 description : String Private	[Is static]
 geometry : Private Alias: 3-07 current geolocation geometry = geolocation = coordinates + elevation	[Is static]
 3-10 Private	

ASSOCIATIONS		
 Association (direction: Unspecified)		
Source: Public is_hosted_by (Class) facility Cardinality: [1]		Target: Public Cardinality:
 Association (direction: Unspecified) 3-07 temporalGeometry		
Source: Public groups (Class) facility Cardinality: [0..*]		Target: Public Cardinality:
 Association (direction: Unspecified) 3-07 temporalGeometry		
Source: Public describes (Class) facility Cardinality: [1]		Target: Public temporalGeom Cardinality:
 Association (direction: Unspecified)		
Source: Public (Class) facility		Target: Public temporalGeom
 Association (direction: Unspecified)		
Source: Public is_hosted_by (Class) facility Cardinality: [1]		Target: Public Cardinality:

facilitySet

Class in package 'WMDR2-core'

ATTRIBUTES		
 id : Private	Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)	[Is static
 manufacturer : Private	This should come from a registry/catalogue, but should accept any organization.	[Is static
 model : Private	Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)	
	This should come from a registry/catalogue, but can accept "custom-built".	[Is static
ASSOCIATIONS		
 Association (direction: Unspecified) 5-09 instrument		
Source: Public describes (Class) deployment		Target: Public
Cardinality: [0..*]		Cardinality

instrument

Class in package 'WMDR2-core'

ATTRIBUTES

Specify using <http://codes.wmo.int/wmdr/Geometry>

[Is static



observedVariableCapability : Private

Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False)

Specify as an array of values taken from <http://codes.wmo.int/wmdr/ObservedVariable> or

[Is static



verticalRangeCapability : Private

Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

Specify using an array of [lower boundary, upper boundary] in meter.

[Is static

ASSOCIATIONS

Association (direction: Unspecified) 5-09 instrument

Source: Public is_used_by (Class) deployment

Cardinality: [1..*]

Target: Public

Cardinality

observation

Class in package 'WMDR2-core'

observation (or observingCapability) wraps one or several deployments used to generate the same observed variable and the same observedGeometryType. Observed variable Cf. 1-01 observedVariable, 1-01 observedGeometryType for more detail.

ATTRIBUTES

definition, human-readable and describing all relevant aspects of an observed variable. It includes the observedProperty, the object of interest, the matrix and domain, as well as constraints. Multiple controlled vocabularies that may be domain-specific.

Examples include:

- observableProperty: state variable (e.g., temperature, pressure, leaf-area-index (LAI), ...), thickness, concentration, scattering coefficient, color, ...)
- objectOfInterest: sea ice, CO2, aerosol particles, chlorophyll
- observablePropertyConstraints: a conventional level above ground (e.g., 2m above ground), a size fraction (e.g., PM2.5), a specific wavelength (e.g., 635 nm), a pressure (range), a temperature, etc., as relevant for the result of the observation. Constraints shall be documented as name and value. Names shall come from a code list, and the values shall come from a code list depending on the name.
- matrix: atmosphere, interstitial air in the root space, air in an ice core, sea surface, soil, ...

These elements will be supported by code lists from codes.wmo.int/wmdr2.

Applications may want to guide users to select a specific observedVariable by offering guided visualizations according to matrix, objectOfInterest, etc.

[Is static]

 1-03 temporalExtent : Private

Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

aggregate derived from deployments if deployments are provided; otherwise, specify beginning and end of observation

[Is static]

 10-01 contact : Private

Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False)

ogc:contact includes "role"

[Is static]

 2-02 programAffiliation : Private

Multiplicity: ([1..*], Allow duplicates: 0, Is ordered: False)

[Is static]

 7-14 internationalExchange : Boolean Private

[Is static]

ASSOCIATIONS		
Cardinality: [1..*]		Cardinality
 Association (direction: Unspecified)		
Source: Public is_hosted_by (Class) facility Cardinality: [1]		Target: Public Cardinality

procedure

Class in package 'WMDR2-core'

- Extension point to document
- ObservingProcedure
 - (Raw)DataProcessing
 - ...

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ASSOCIATIONS		
 Association (direction: Unspecified)		
Source: Public describes (Class) deployment Cardinality: [1..*]		Target: Public Cardinality
		ObservingProc

temporalGeometry

Class in package 'WMDR2-core'

ATTRIBUTES

ASSOCIATIONS

 Association (direction: Unspecified) 3-07 temporalGeometry

Source: Public describes (Class) facility
Cardinality: [1]

Target: Public
temporalGeom
Cardinality

temporalReportingSchedule

Class in package 'WMDR2-core'

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ATTRIBUTES

 id : Private
Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

[Is static

 interval : Private

An array of arrays of begin(inclusive) and end(exclusive) datetime stamps. If the end is n
open-ended. Intervals can overlap to indicate multiple schedules operating in parallel.

[Is static

 schedule : Private

An array of arrays, whereby each nested array corresponds to an interval and describes a
follows:

ASSOCIATIONS		
	Association (direction: Unspecified)	
Source: Public (Class) facility		Target: Public temporalGeom